SECTION 6 - 1

POWER CONVERSION MODULE

1000 HP AT 600VAC

DS3820PCD

(GEK - 83846)

SECTION 5

HOW THE AC GENERATOR CONTROL WORKS

TABLE OF CONTENTS

SECTION	TITLE	PAGE
5-1	AC Generator/Engine Module (GEM)	5 - 1
5-1.1	Function	5 - 1
5-1.2	Theory of Operation	5 - 1
5-1.2.1	Governor Control	5 - 1
5-1.2.2	Voltage Regulation	5 - 1
5-1.2.3	Metering	5 - 2
5-1.2.4	Fault Detection	5 - 2
5-1.3	Removal and Replacement Procedures	5 - 4
5-1.4	Elementary Diagrams	5 - 5
5-1.4.1	Flow Diagram Explanations	5 - 7
5-1.4.2	Signal Cross Referencing	5 - 8
5-2	KVA/KW Module	5 - 20
5-2.1	Function	5 - 20
5-2.2	Theory of Operation	5 - 20
5-2.2.1	KW Limit	5 - 20
5-2.2.2	KVA Limit	5 - 20
5-2.2.3	High Load Indication	5 - 21
5-2.2.4	KW Totalizing	5 - 21
5-2.3	Module Connections	5 - 21
5-2.4	Elementary Mnemonics	5 - 22
5-2.4.1	Devices	5 - 22
5-2.4.2	Signal Designations	5 - 22
5-2.5	Removal and Replacement Procedures	5 - 24
5-2.5.1	Removal	5 - 24
5-2.5.2	Replacement	5 - 24

TABLE OF CONTENTS (Continued)

SECTION	TITLE	PAGE
5-2.6	Elementary Diagram	5 - 24
5-3	Ground Detector Module	5 - 25
5-3.1	Function	5 - 25
5-3.2	Removal and Replacement Procedures	5 - 25
5-3.2.1	Removal	5 - 25
5-3.2.2	Replacement	5 - 25
5-3.3	Elementary Diagram	5 - 25

LIST OF ILLUSTRATIONS

NUMBER	DESCRIPTION	PAGE
Figure 5	Flow Diagrams	5 - 7
Figure 5 - 1	Operation of GEM Engine Governor	5 - 9
Figure 5 - 2	Operation of GEM Voltage Regulator and Power Supply	5 - 10
Figure 5 - 3	Operation of GEM Metering	5 - 11
Figure 5 - 4	Operation of GEM Circuitry	5 - 12
Figure 5 - 5	Elementaries	
	Generator Electronics Module (480VAC) (173C5579, Sheets 4A - 4D)	5 - 13
	AC Generator Control Module Interconnections (DS3820WGC, Sheets 4AA - 4DA)	5 - 17

5-1 AC GENERATOR/ENGINE MODULE (GEM)

5-1.1 Function

The AC Generator Engine Module (GEM) is responsible for starting the engine, maintaining constant engine speed, and regulating the generator output voltage.

5-1.2 Theory of Operation

The GEM is divided into four functional areas:

1. governor control
2. voltage regulation
3. metering
4. fault detection.

Each of these areas is discussed below and illustrated by flow diagrams on the following pages. A flow diagram is a condensed, easy-to-read summary of the operation of a circuit or module. If you are not familiar with the symbols used in flow diagrams, be sure to read Section 5-1.4.1, titled "Flow Diagrams" before reading the next four functional descriptions.

5-1.2.1 GOVERNOR CONTROL (See Figure 5-1) - The GEM governs the engine speed by controlling the engine actuator (throttle) position with an electrical signal. The Engine Control Selector on the front of the GEM is involved in engine speed control and has four positions: OFF, IDLE, RUN and LOAD. In the OFF position, the engine is shut down. In the IDLE position the engine can be started and will run at idle speed. In the RUN position, the engine runs at normal speed but the generator will not pick up or share load with other on-line generators. In the LOAD position, the engine runs at normal speed and the generator will pick up load and work with other on-line generators to equally share the load.

5-1.2.2 VOLTAGE REGULATION (See Figure 5-2) - The GEM will maintain the generator output voltage at a steady value under changing loads. This is accomplished by comparing the generator output voltage with a reference and increasing or decreasing the generator field voltage, which controls the output voltage, to minimize any difference.

The Frequency Trim Adjustment located below the Engine Control Selector is used to synchronize the frequency of the generator being brought on line to the AC bus frequency.

5-1.2 Theory of Operation (Continued)

5-1.2.3 METERING (See Figure 5-3) - There are four external meters that monitor the generator output:

Kilowatt Meter - Indicates how much power is being generated by the engine. Used to ensure that the engines are all sharing the load.

Kilovar Meter - Indicates the reactive power delivered by the generator. Used to ensure that the generators are properly paralleled.

Voltmeter - Indicates the voltage of the source selected by the four position Voltmeter Switch on the GEM front panel. The first three positions of the switch select the line to line voltage of the generator to be monitored. The fourth position selects the AC bus voltage to be monitored.

Ammeter - Indicates the line current for the phase selected by the Amps Switch on the GEM front panel.

5-1.2.4 FAULT DETECTION (See Figure 5-4) - One red and six yellow indicating lights on the front panel of the GEM annunciate fault conditions. Under normal operating conditions with the generator on-line, all lights (yellow and red) will be on. The first fault to occur causes its associated light to flash and subsequent faults cause the associated lights to go out. Pressing the reset pushbutton will reset the control, but if a fault still exists the appropriate light will flash and/or go out. All faults must be cleared before the generator breaker can be closed. The fault lights are listed and described below.

Gen On (red) - When on, indicates that the generator breaker is closed and the generator is supplying power to the AC bus. When off, indicates that the generator is not supplying power to the AC bus. Most of the following faults will trip the generator breaker and cause the "Gen On" light to go out.

Undervolt (yellow) - When off, indicates that a generator undervoltage condition exists. This fault will trip the generator circuit breaker.

5-1.2.4 FAULT DETECTION (Continued)

No OV/UF (yellow) - When off, indicates that a generator overvoltage or underfrequency condition exists. The control will trip the generator circuit breaker for an overvoltage condition, but not for an underfrequency condition. If the overvoltage condition still exists twenty seconds after the breaker has been tripped, the control will shut down the engine.

No Rev Pwr (yellow) - When off, indicates that a reverse power condition has been detected. This means that power from other generators is flowing into the affected generator. This fault will trip the generator circuit breaker.

No Ovrspd (yellow) - When off, indicates that a generator engine overspeed condition exists. This fault will trip the generator circuit breaker and shut down the engine.

No Tachometer Loss (yellow) - When off, indicates that the tachometer signal from the engine has been lost. This fault will trip the generator circuit breaker and shut down the engine.

Pwr Supply OK (yellow) - When off, indicates that the GEM power supply has malfunctioned. This fault will trip the generator circuit breaker and shut down the engine.

5-1.3 Removal and Replacement Procedures

5-1.3.1 GENERATOR CONTROL MODULE (GEMA) -

Removal: **Disconnect all power to the panel.** Note the location of plugs to be disconnected: JXAC, JSYSA, JSYSB, JSYSC, JPT, JDC1, JDC2, and JDC3. Also note the order of connections at the GEMC terminal board at the top of the module. Disconnect all of these wires and plugs.

Loosen the 6 GEMA mounting screws (3 on each side of the module) two full turns each. Grasp the module at the steel sides covering the card rack. Lift upward and toward the front of the panel to withdraw the mounting screws through the teardrop mounting holes. Remove the module from the panel.

Replacement: Mount the replacement module by reversing the order of the above steps.

5-1.3.2 GENERATOR CONTROL TRANSFORMER MODULE (GEMB) -

Removal: **Disconnect all power to the panel.** Remove the GEMA module according to the removal instructions above.

Loosen the 4 GEMB mounting screws (2 at the top and 2 at the bottom) two full turns each. Firmly grasp the framework and lift upward and toward the front of the panel to withdraw the mounting screws through the teardrop holes at the top of the module and the slots at the bottom. Remove the module from the panel and place it upright on a table beside the replacement module.

Replacement: Facing the rear of the modules, ensure that the jumpers on the transformer terminal board are connected **exactly** the same on the replacement module as on the original module. Later versions of this module have a nameplate located on the sideplate which identifies the jumper connections for a given generator field voltage.

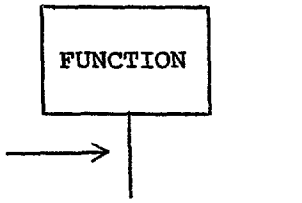
Replace the module by reversing the order of the above removal procedure.

5-1.4 Elementary Diagrams

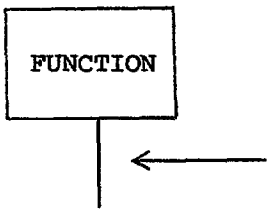
The remainder of this tab section contains flow charts and elementary diagrams for the AC generator/engine module.

NOTES

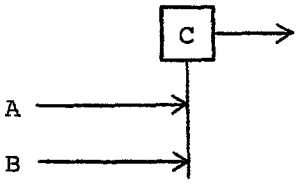
5-1.4.1 FLOW DIAGRAMS - The signal flow diagrams on the following pages use the symbols below to help explain each function:



A signal terminating on the LEFT of the center line is always a permissive to start and maintain the function once it is started. If any one of these permissives is lost, the function becomes inactive.



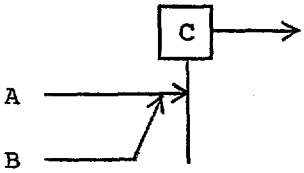
A signal terminating on the RIGHT of the center line is required only at the start of the function. After starting, if the signal is lost, the function remains active.



This is a logical AND function.

A · B = C

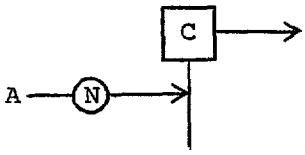
A AND B cause function C.



This is a logical OR function.

A + B = C

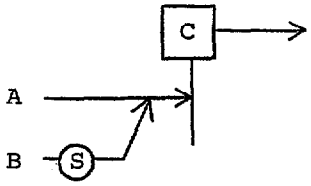
A OR B cause function C.



This is a logical NOT function.

$\overline{A} = C$

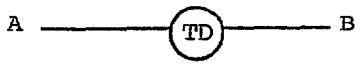
If a signal is NOT present, then function C is active.



This is a conditioned OR or "seal in" function.

A = C

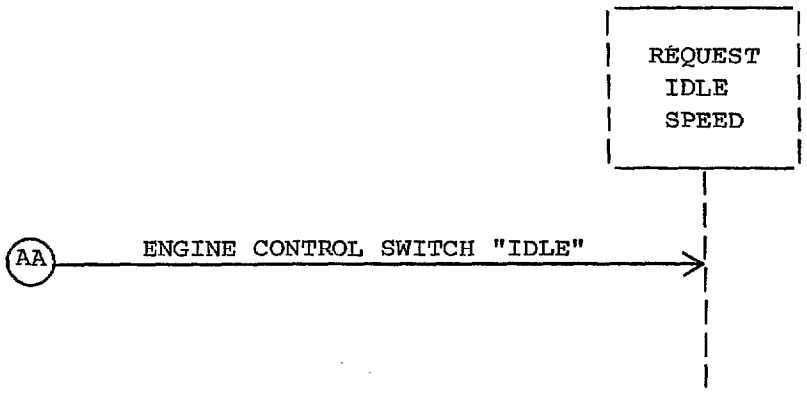
A causes function C, which is then "sealed" by B.



This is a time delay function.

Figure 5: Flow Diagrams

5-1.4.2 SIGNAL CROSS REFERENCING - Signals are cross-referenced from sheet to sheet by being labeled with a sheet identification (A, B, C, or D, which specifies on which sheet the signal originates) and a label, within brackets, (A, B, C, etc.,) which differentiates between several signals on a sheet. In the following example, the signal ENGINE CONTROL SWITCH "IDLE" is the first one used on sheet A, hence its identification as AA. The next signal used would be AB, and so forth. This signal is shared with other points cross-referenced in the circle or brackets. If more than one point shows AA, it only means that all those points share the same signal.



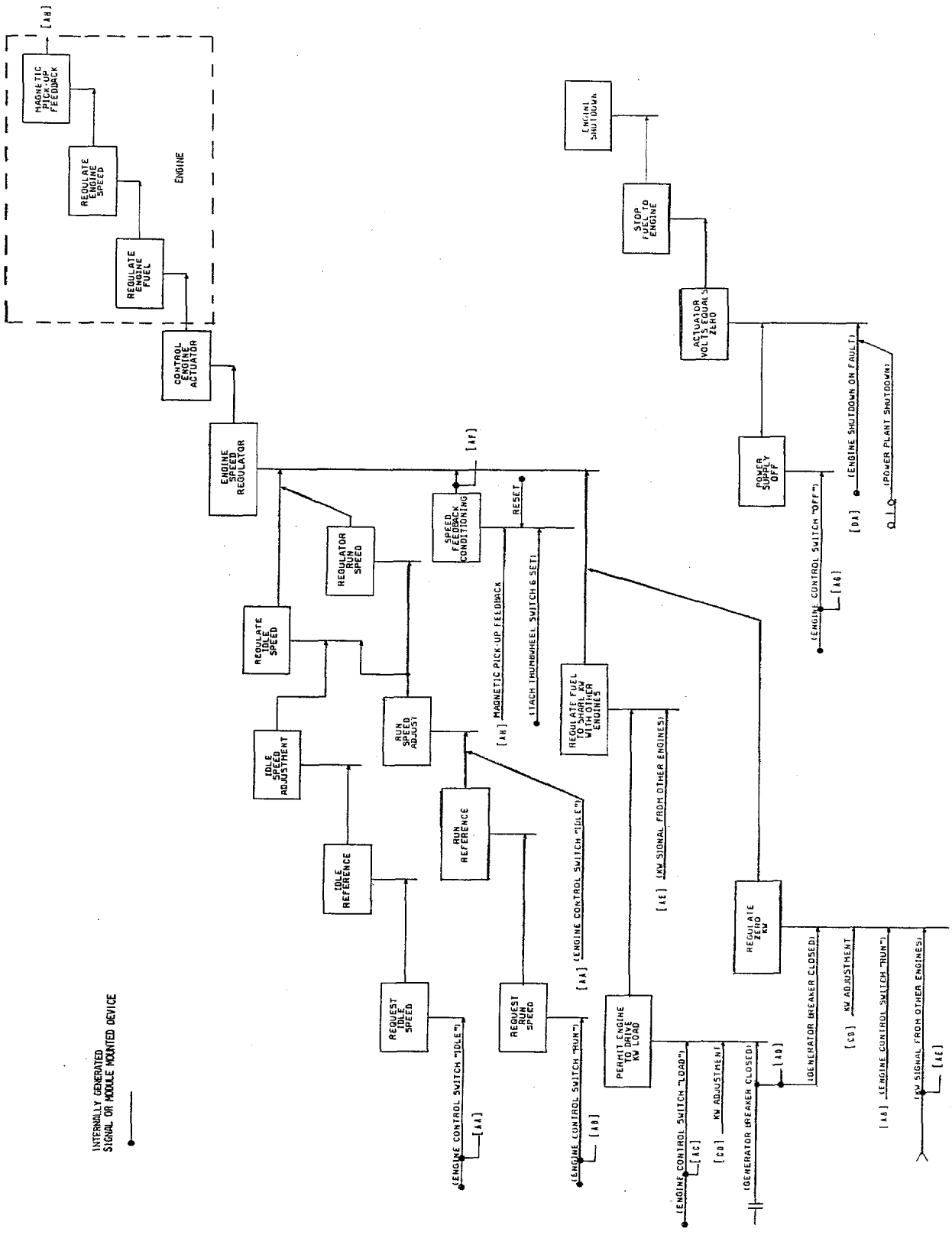


FIGURE 5-1: OPERATION OF GEM ENGINE GOVERNOR

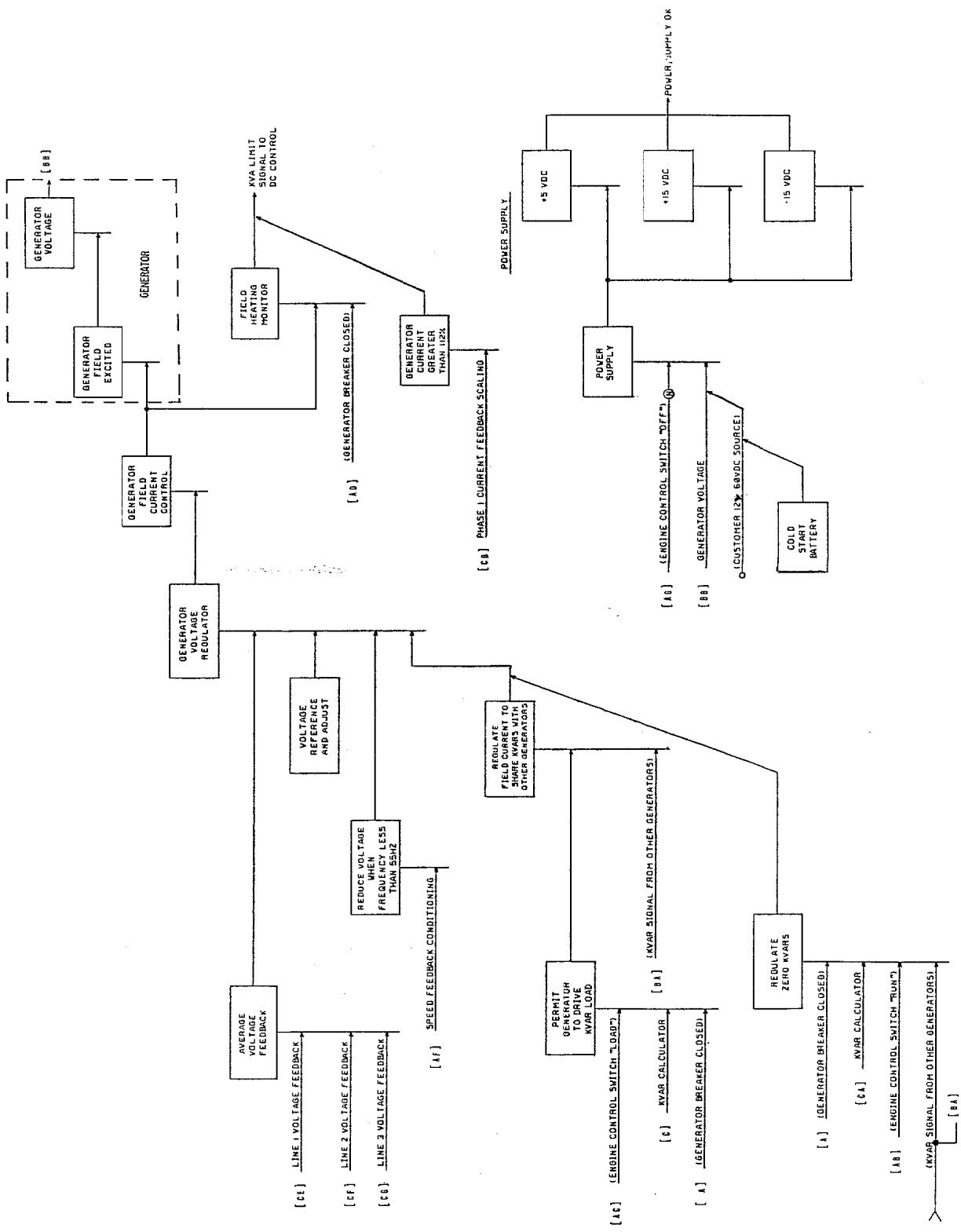


FIGURE 5-2: OPERATION OF GEM VOLTAGE REGULATOR & POWER SUPPLY

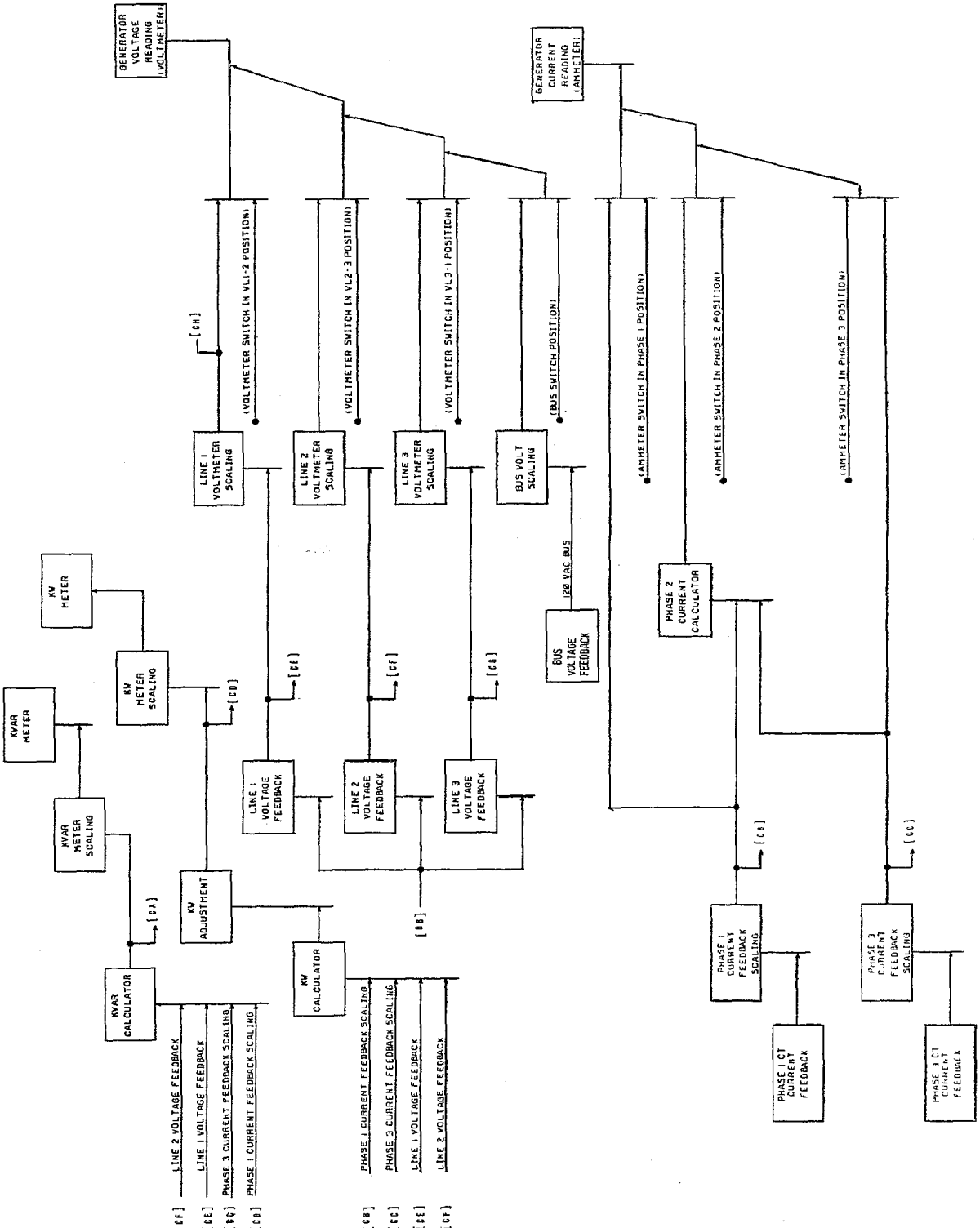


FIGURE 5-3: OPERATION OF GEM METERING

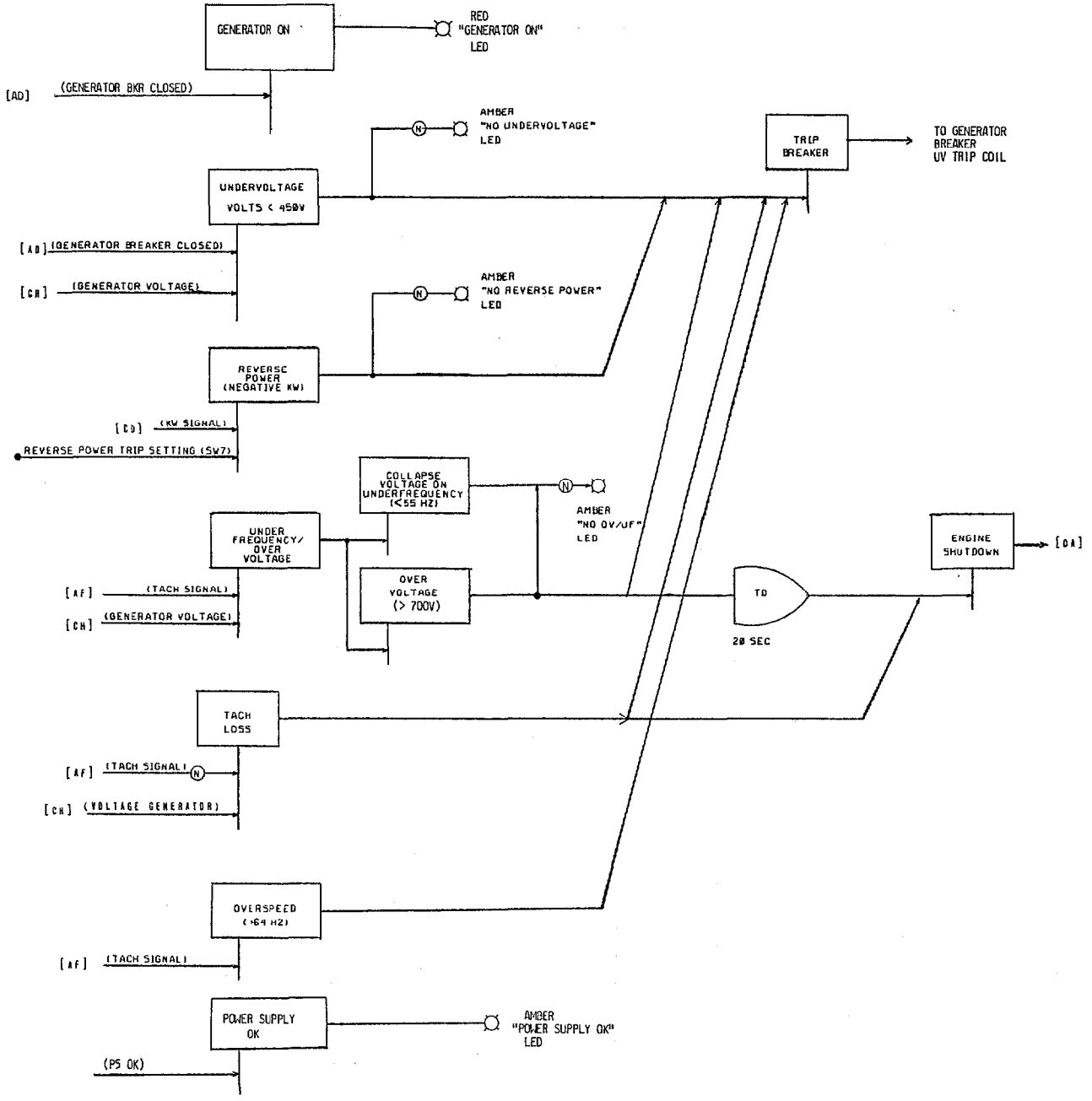
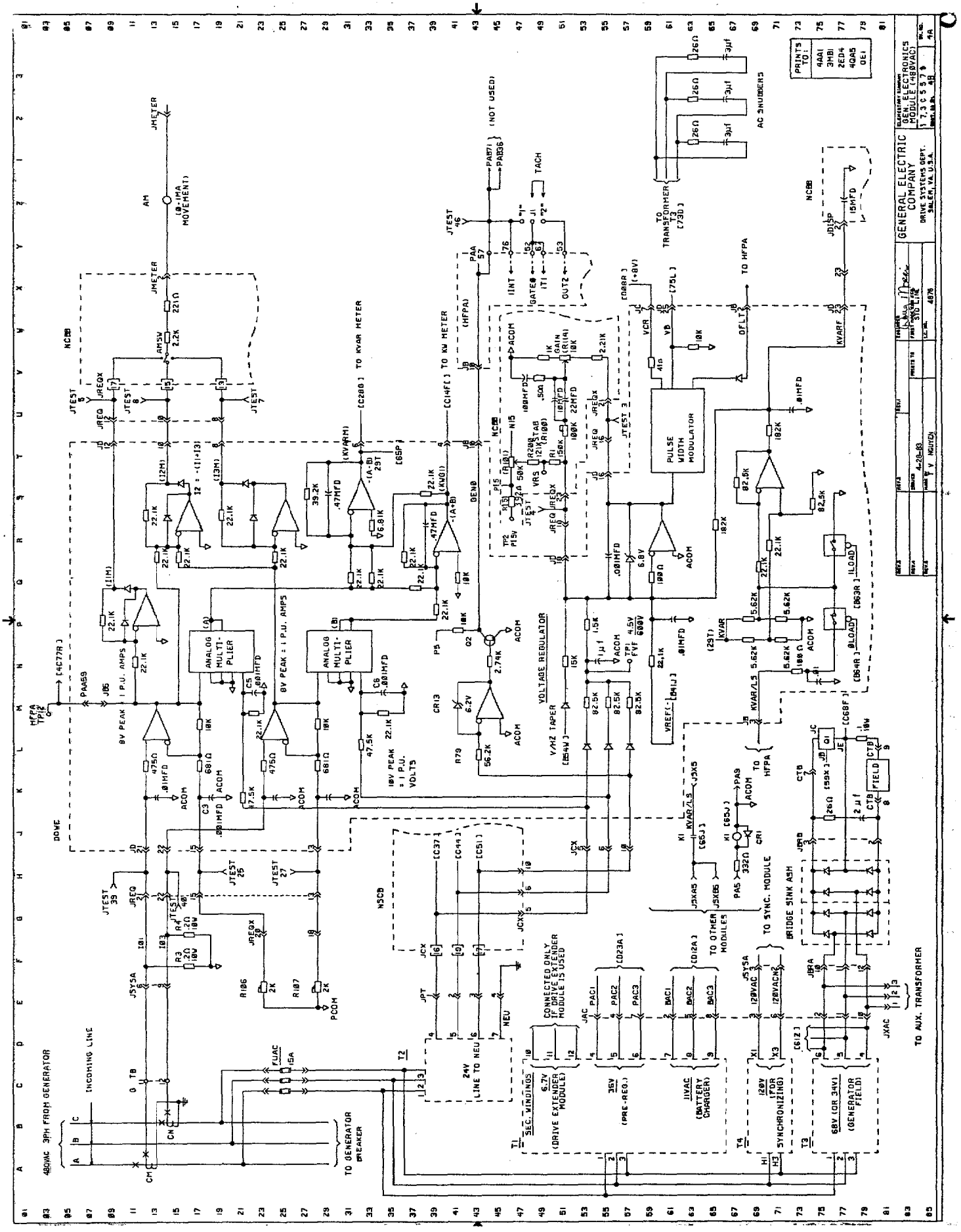
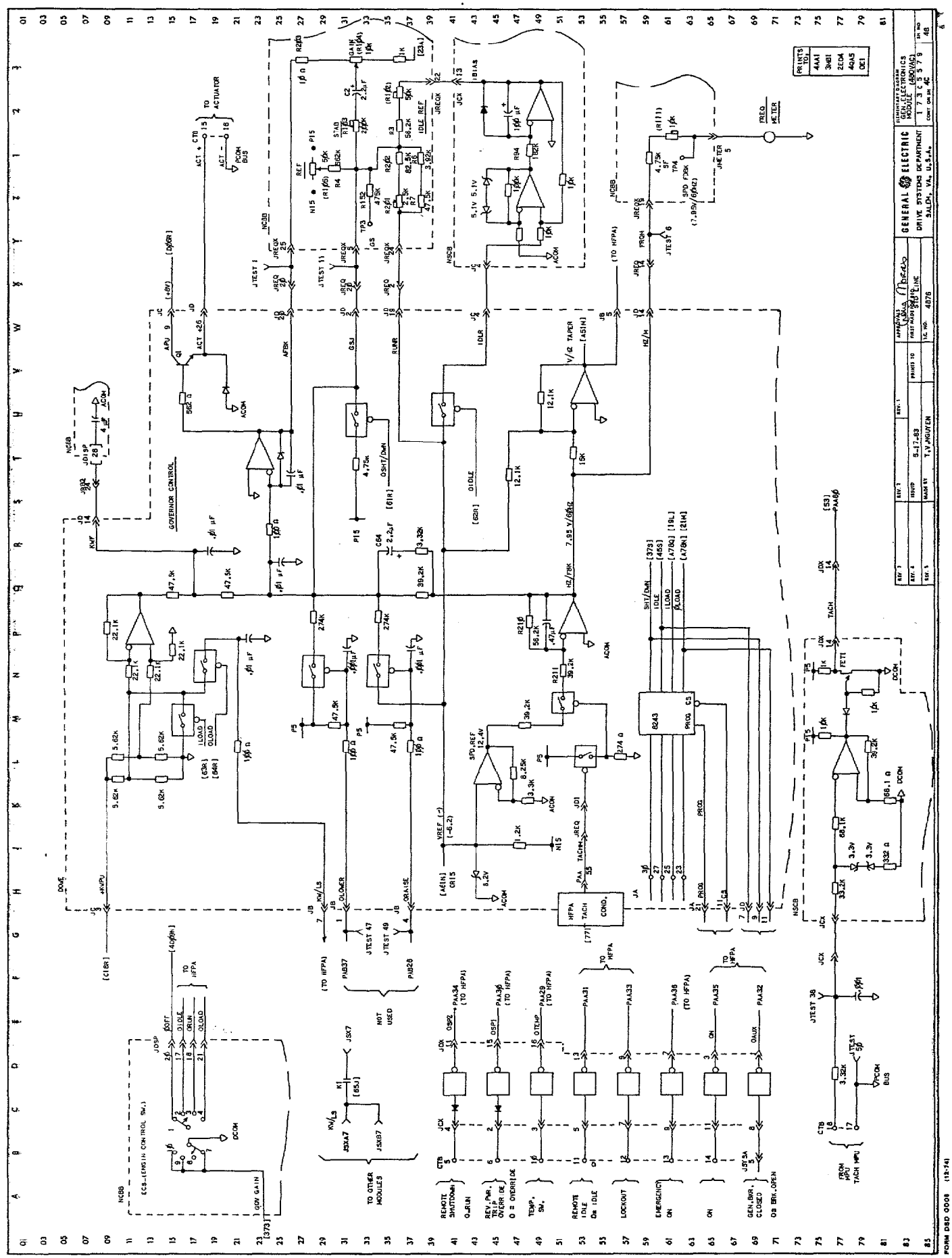
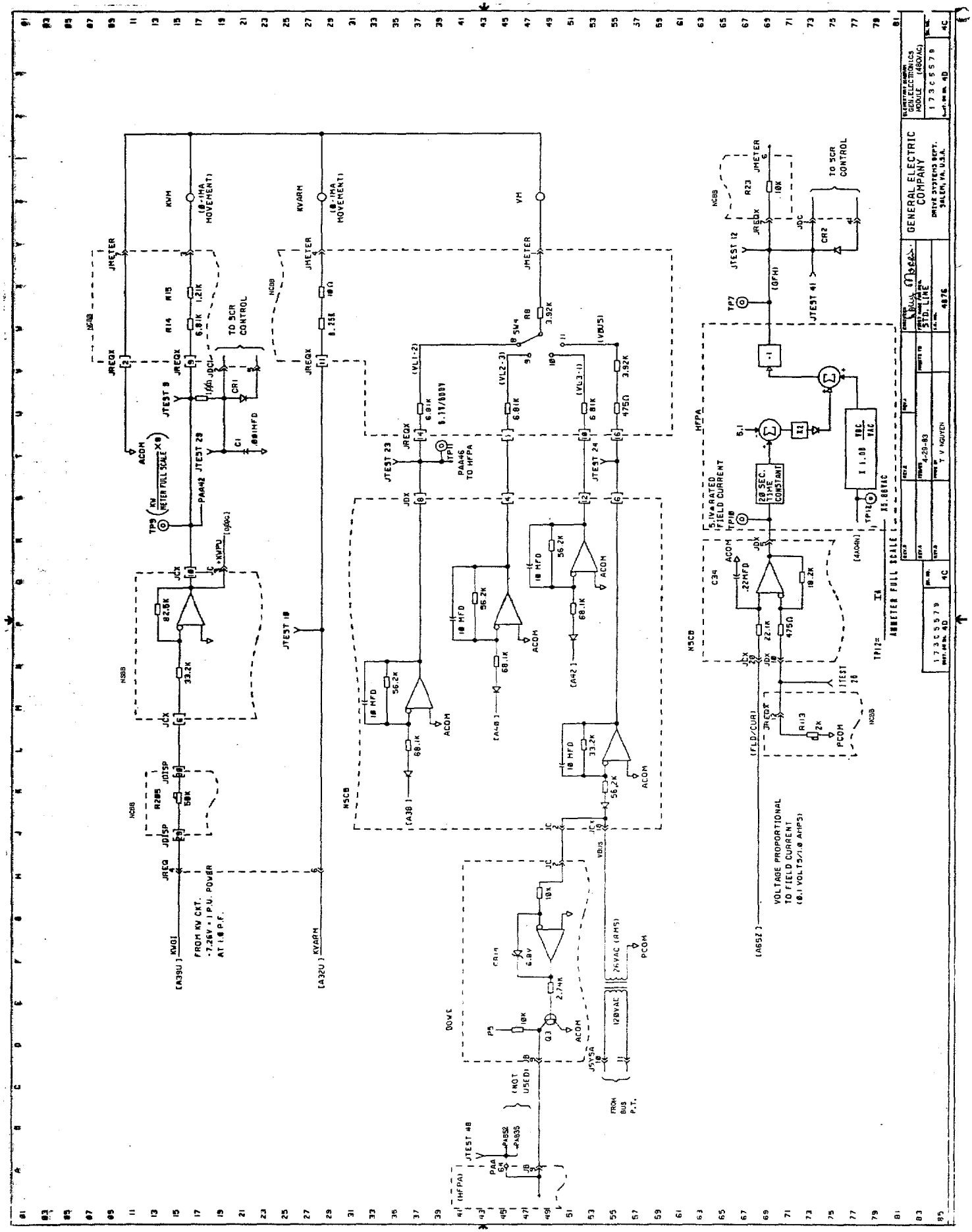
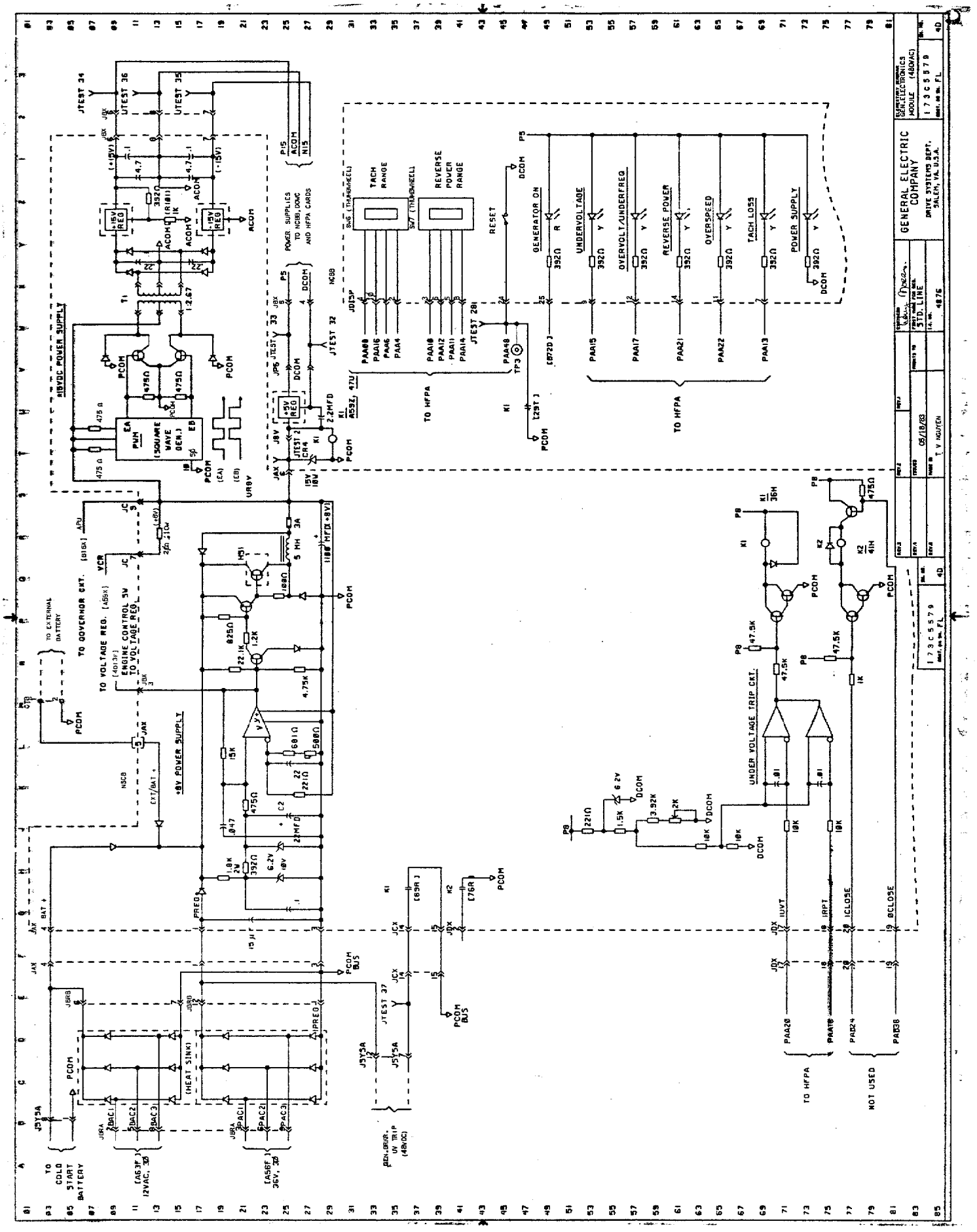


FIGURE 5 - 4: OPERATION OF GEM CIRCUITRY

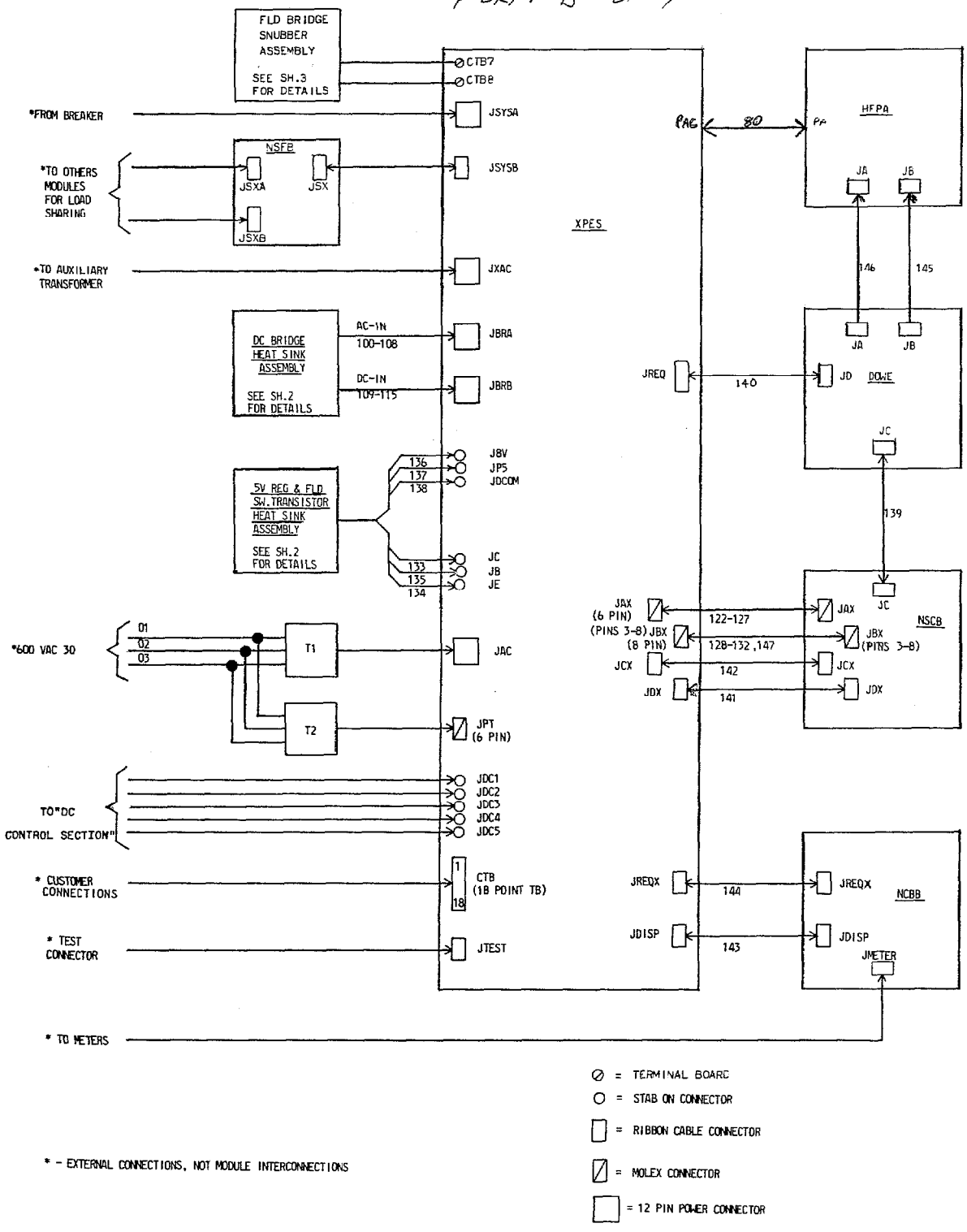

GENERAL ELECTRIC COMPANY
SALEM, VA. U.S.A.
DRIVE SYSTEMS DEPT.
MODULE (480VAC)
480VAC 3PH FROM GENERATOR
INCOMING LINE
TO GENERATOR BREAKER
TO AUX. TRANSFORMER
TO SYNC. MODULE
TO OTHER MODULES
BRIDGE SINK ASH
AC SNUBBERS
TO TRANSFORMER T2 (730)
TO HFPA
TACH
AM
(0-1MA MOVEMENT)
JMETER
NCBB
NSCB
DOVE
ANALOG MULTI-PLIER
8V PEAK = 1 P.U. AMPS
8V PEAK = 1 P.U. VOLTS
TO KW METER (HFPA)
TO KVAR METER
VOLTAGE REGULATOR
PULSE WIDTH MODULATOR
CONNECTED ONLY IF DRIVE EXTENDER MODULE IS USED
24V LINE TO NEU
6.7V SEC. WINDINGS (DRIVE EXTENDER MODULE)
35V (PRE-REG.)
115VAC (BATTERY CHARGER)
120V (FOR SYNCHRONIZING)
68V (FOR 34VI (GENERATOR FIELD)
T1
T2
T3
T4
PRINTS TO:
4AA1
3MB1
2ED4
4QA5
0E1






GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
1736557 9
40
05/18/88
T.V. NGUYEN
STD. LINE
4875
SCHEMATIC DIAGRAM GEN. ELECTRONICS MODULE (480VAC)
+8V POWER SUPPLY
+15VDC POWER SUPPLY
TO VOLTAGE REG. (45H)
TO GOVERNOR CKT. (81K)
TO EXTERNAL BATTERY
TO COLD START BATTERY
UNDER VOLTAGE TRIP CKT.
TO HFPA
NOT USED
PCOM BUS
(HEAT SINK)
POWER SUPPLIES TO NCBB, DOAG AND HFPA CARDS
TACH RANGE
REVERSE POWER RANGE
RESET
GENERATOR ON
UNDERVOLTAGE
OVERVOLT/UNDERFREQ
REVERSE POWER
OVERSPEED
TACH LOSS
POWER SUPPLY
PULSE WIDTH MODULATED SQUARE WAVE GEN.
PWM (SQUARE WAVE GEN.)
EA
EB
PCOM
DCOM
ACOM
NCBB
NCSB
PREG
VR8V
VCR
EXT/BAT +
IUVT
IRPT
ICLOSE
BCLOSE
PAA20
PAA18
PAB24
PAB38

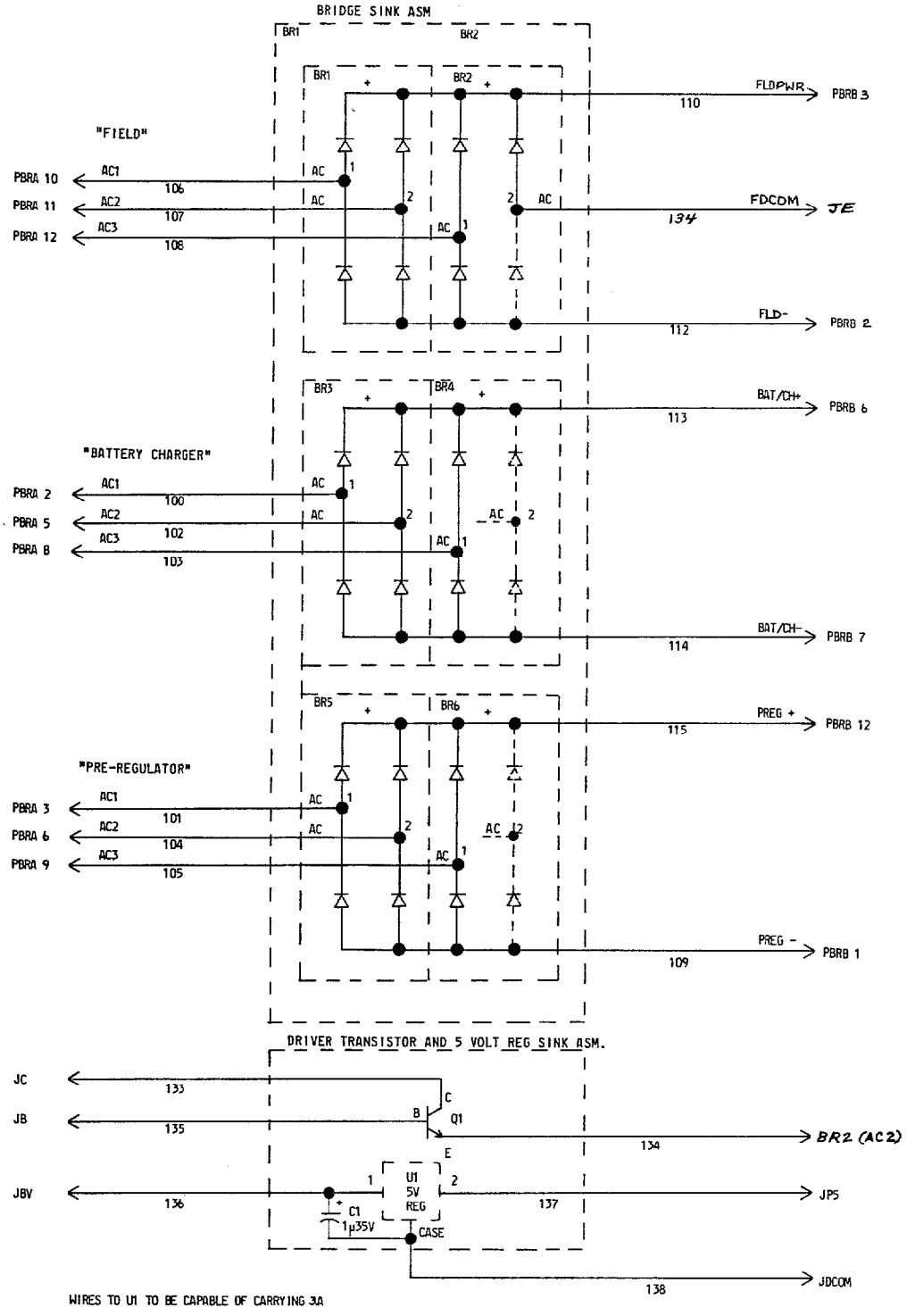
FORM "B" ONLY



REV. 1 R.L. 3/5/81 OEH	REV. 4	REV. 5	PRINTS TO DL114	ENGINEER Gary Stith	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM AC GENERATOR CONTROL MODULE INTERCONNECTIONS
REV. 2 OEK CEM 7/7/81	ISSUED 6/13/80					
REV. 3 OEG 3/19/82 WCT	MADE BY R.E.VANDERPOOL					DS3820WGC CONT. ON SH. 4BA SH. NO. 4AA

CAR

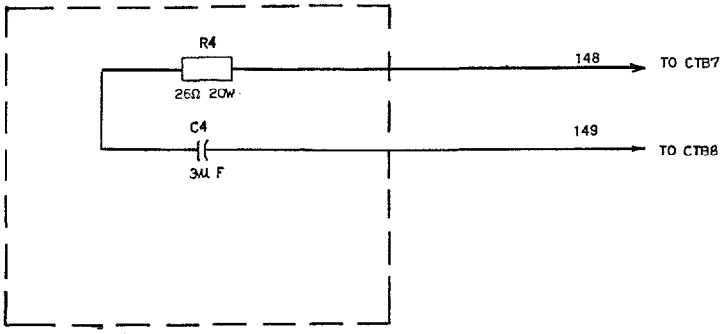
FORM "B" ONLY



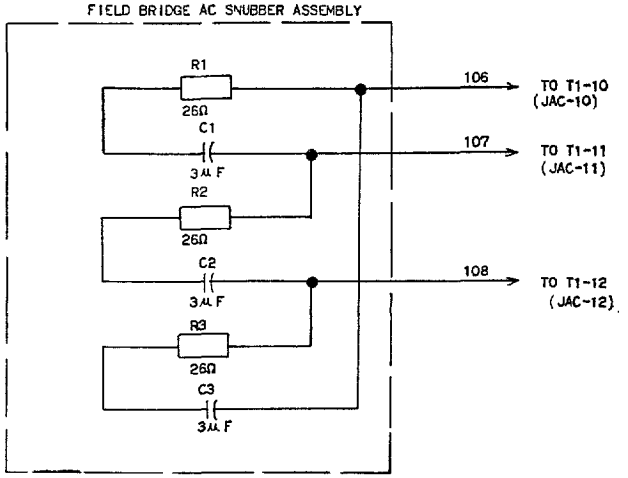
REV.1 OEG 1/28/81 HC	REV.4 OEM 12/19/81 HC	REV.5	PRINTS TO DL114	ENGINEER	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV.2 R.L. 3/5/81 OEH	ISSUED 6/13/80				DRIVE SYSTEMS DEPARTMENT	AC GENERATOR CONTROL MODULE
REV.3 OEK 7-2-81 CCM	MADE BY R.E. VANDERPOOL				SALEM, VA. U.S.A.	DS3820WGC
						CONT. ON SH 4CA SH. NO. 4BA

FORM "B" ONLY

FLD BRIDGE SNUBBER ASSEMBLY



FIELD BRIDGE AC SNUBBER ASSEMBLY



REV. 1 OGC-2 4-6-83 WCD	REV. 4	REV. 5	PRINTS TO CL114	ENGINEER D.E. Lee 8/07/82	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM OWD AC GENERATOR CONTROL MODULE
REV. 2 OGK 10-19-84 WCD	ISSUED 7/2/82					DS3820WGC
REV. 3	MADE BY D.E.GEE					CONT. ON SH. 4A · SH. NO. 4CA

5-2 KVA/KW MODULE

5-2.1 Function

The KVA/KW module consists of one (1) nine slot row containing printed circuit cards and terminal boards. This module is hinged and mounted in a control section for the generators. It contains the KVA and KW limiting and alarm circuits.

5-2.2 Theory of Operation

The KW/KVA circuitry performs four separate functions as described below.

5-2.2.1 KW LIMIT - The purpose of the KW (kilowatt) limit circuit is to prevent overloading of the diesel engines. It does this by providing a signal proportional to the magnitude of the KW overload. This signal is sent to all power converter regulators which then act to reduce converter output, thereby reducing engine loading.

External KW sensors monitor the KW being supplied by each generator. The **highest** KW signal of all the generators on-line is selected and supplied to the input of inverting amplifier AOAL(A). This signal is compared to a preset bias signal from trimpot P1. If the (+) KW signal exceeds the (-) bias signal, amplifier (A) output will swing negative. This negative signal is supplied to all regulators by means of ribbon connector JKW - Slot 0H, and reduces converter output to the level required to prevent engine overload. A clamping diode, CR7, prevents amplifier AOAL(A) from swinging positive. Capacitor C1 provides a lagging time constant to help stabilize the system.

5-2.2.2 KVA LIMIT - The purpose of the KVA limit circuit is to protect the AC generator from overload. It does this in a manner similar to that for KW limit described above. Amplifier (D) accepts the **highest** KVA input via inverting amplifier AOAL(C) and trimpot P3 sets the desired limit. The KW limit and the KVA limit signals are diode OR'ed (diodes CR1 and CR2). This ensures that either an excessive KVA load or an excessive KW load will reduce the power converter output.

5-2.2 Theory of Operation (Continued)

5-2.2.3 HIGH LOAD INDICATION - The purpose of this circuit is to indicate by means of a light on the driller's console whether the system is approaching or is in KW or KVA limit. The negative KW limit signal is inverted by amplifier AOAL(E) and applied to a gate unit AOAL(G). When this input becomes positive (IN 0V) the output will go to +5 volts. This signal is applied to amplifier AOAL(F) - a high gain, non-inverting power amplifier. Its output will swing negative to energize reed relay HLR whose interlock is used to power the HIGH-LOAD light on the console.

Similarly, when a KVA limit drives amplifier AOAL(D) negative, gate unit AOAL(H) is triggered. This unit delivers a +5 volt output when its input goes negative, (in 0V), again driving AOAL(F) output negative to energize relay HLR.

5-2.2.4 KW TOTALIZING - Individual signals proportional to each generator KW load are summed in amplifier AOAL(B). It's output drives the "% KW" meter on the console. The meter is divided into segments - one segment for each of the available engines - and calibrated (trimpot P2) so that each segment represents 100% KW load per engine. This allows the attendant to judge the degree of loading versus the number of engines on line.

5-2.3 Module Connections

All voltage levels on the AC interface module are +28 volts. All like wire numbers are connected by back plane wiring from point to point on the module.

Each module terminal board and connector is described below.

5-2.3.1 TBKW (SLOT 0E) - 12-point terminal board for random hard wiring (if used).

5-2.3.2 JKW (SLOT 0H) - card with four (4) 10-pin connectors. Only the bottom connector is wired. KVA/KW limit signals are sent back and forth between the AC interface module and the power converter panels by means of a 10-conductor ribbon cable connected continuously panel-to-panel.

5-2.4 Elementary Mnemonics

The following mnemonics or abbreviations are used in the elementary shipped with your equipment. Their meanings are explained below.

5-2.4.1 DEVICES -

C ___ - Identifies a Capacitor

CR ___ - Identifies a Diode

DC - Driller's Console 51-pin receptacle

HLR - Identifies a High Load Relay

JKW - 10-pin Jack for KW signal output

P ___ - Identifies a Potentiometer

R ___ - Identifies a Resistor

TBKW - 12-Point Terminal Board for KW signal output

5-2.4.2 SIGNAL DESIGNATIONS -

ACOM - Analog Common. Common connection for signals input to the module (except for high level signals)

ACOM-GC - Analog Common - Generator Control. Common connection for signals input to module from the generator/engine module

ALARM - Alarm Signal. Activates alarm horn on operator's console

AMCOM - Alarm Signal Common. Common for signal to activate alarm horn on operator's console

BATCOM - 12 Volt Battery Common Lead

BKRCLS# - Generator number # Main breaker close signal from the synchronizing circuit

BUSCOM - Generator Control 120 volt AC Bus Common

DKVA - Diode or KVA signal indicating highest value of generator output. (also see KVA - signal designation)

5-2.4.2 SIGNAL DESIGNATIONS - (Continued)

DKW - Diode blocked KW signal from generators which represents the largest output being generated. (also see KW + signal designation)

GEN # KW - KW Limit signal for Generator number # KW. (also see KW - # signal designation)

GENLT # - Generator Light signal to "GEN ON" light on operator's console for generator number #

HLD - High Load signal to indicating light on the operator's console

KVA - Diode or KVA signal indicating highest value of generator output (also see DKVA signal designation)

KVA/KW - Kilovolt-Amperes/KiloWatt limit signal to PCP interface module

KW-# - Generator number # KW limit signal (also see GEN # KW signal designation)

KW + - Diode Blocked KW signal from generators which represents the largest output being generated (also see DKW signal designation)

KW - Clamped negative KW limit signal

LFLT - Low fuel light signal to indicating light on operator's console

N15 - Negative 15 volt DC bus

OVRTP - PCR Overtemperature signal to indicating light on operator's console

PCKW - Percent KW signal to meter on operator's console

PCOM - Peripheral Common

PXB1 - Positive lead of customer external battery

P15 - Positive 15 volt DC bus

P12V - Positive 12 Volt DC bus

RSTD1 - Remote Shutdown signal from remote customer operator

SIGCOM - AC Generator control panel low level signal common

5-2.4.2 SIGNAL DESIGNATIONS - (Continued)

SYNCOM - Synchronizing common. Common for signals used in generator synchronization.

UVP - Undervoltage Trip Signal (Positive), from emergency shutdown pushbutton on operator's console.

VBUS - Control instrument potential transformer Bus Voltage signal.

VCLS-# - Generator number # synchronizing circuit voltage signal to close

VGEN-# - Generator number # incoming phase 2 generator voltage

% KW - Percent KW signal to meter on operator's console (also see PCKW signal designation)

24VAC - 24 volt AC bus

24VCOM - 24 volt AC bus common

5-2.5 Removal and Replacement Procedures

5-2.5.1 REMOVAL - **Remove all power from the panel and module.** Ensure that sufficient information is available to help you reconnect the module after removal. See the elementary drawing in this section for connection details.

Remove all connections to the face of the module.

Remove wires attached to terminal boards TBKW (slot 0E).

Remove the 2 bolts at the left of the module and gently swing the module to the right side.

While holding the module in position, remove the 2 bolts under the hinge on the right side and remove the module from the panel.

5-2.5.2 REPLACEMENT - Install the new module by reversing the above removal procedure.

5-2.6 Elementary Diagrams

See elementary diagram, 229B3724 - Sh. 23B - D, for optional KVA/KW module with the POWER BOOST supplied on this requisition.

5-3 GROUND DETECTOR MODULE

5-3.1 Function

The Ground Detection Module (GDM) detects AC or DC line-to-ground faults. The GDM contains five indicator lights and a test pushbutton. The top three lights indicate a ground fault on the incoming AC line. There is one light for each phase. If the light goes out, a ground fault has occurred on that phase. The bottom two lights indicate a ground fault on the SCR DC bus. In normal operation both lights should be out. If either the positive or negative DC bus of any SCR package becomes grounded, the appropriate light will come on. The test pushbutton allows you to check the operation of all five lights. Pressing the test pushbutton causes all lights to come on.

Figure 5 - 5 is a diagram of the Ground Detector Module circuitry.

5-3.2 Removal and Replacement Procedures

5-3.2.1 REMOVAL -

- Remove all power to the module by removing the fuses that feed the module.
- See Figure 5-5 for electrical details.
- Unplug the single Mate-N-Lock plug to the module and remove the 4 screws which hold the module in place. Remove the module.

5-3.2.2 REPLACEMENT -

- With all power to the panel disconnected, replace the module by reversing the above removal procedure.

5-3.3 Elementary

See Elementary 229B3724 - Sh. 23AA, for optional ground detector with metering supplied on this requisition.

NOTES